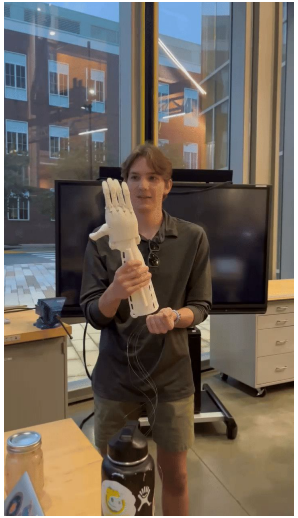
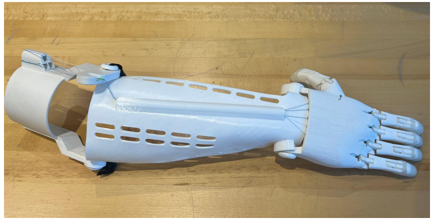
Introduction to CAD using Onshape

By The Accessible Prosthetics Initiative

What’s API?

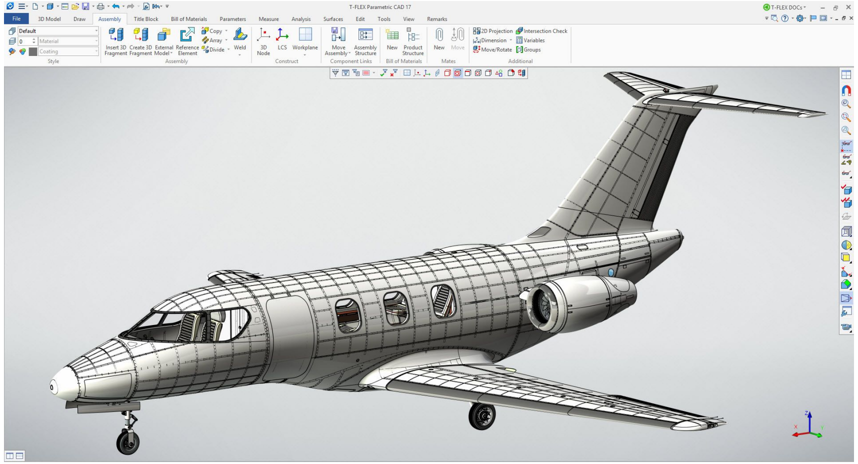
- Accessible prosthetics
  - Goal: inexpensive prosthetic devices
- Helping others
- Teams:
  - Communications
  - Education
  - Production
    - Engineering (all disciplines)
    - Computer Science
  - Business
- Chapters at other institutions
  - Ohio State
  - Brown University
  - Georgia Tech





What is CAD?

- Canadian Dollar
- Center for Agricultural Development
- Computer Aided Design
    - This is what we will be focusing on today.



Examples of CAD Software:

We are using **Onshape** today!

AUTODESK® INVENTOR®

AUTODESK® FUSION 360™

TINKERCAD

AUTODESK® TINKERCAD®

SOLIDWORKS

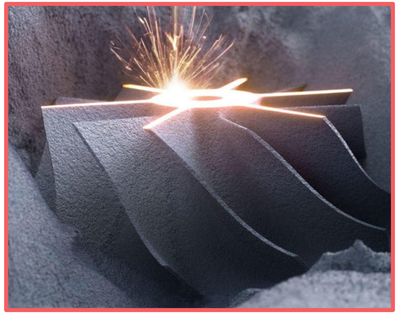
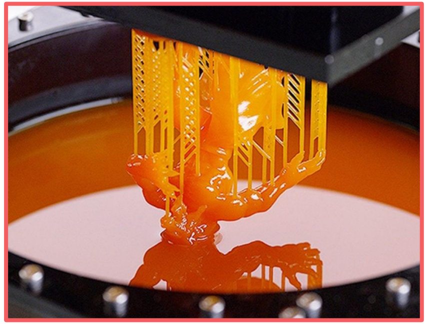
blender®

Onshape

Welcome To The Future Of CAD

Manufacturing

- Manufacturing can be additive or subtractive
  - Additive = adding material to build up the shape
    - Stereolithography
    - Selective laser sintering
    - Fused deposition modeling
  - Subtractive = removing material to carve out the shape
    - Lathing
    - CNC Milling
    - Laser Cutting

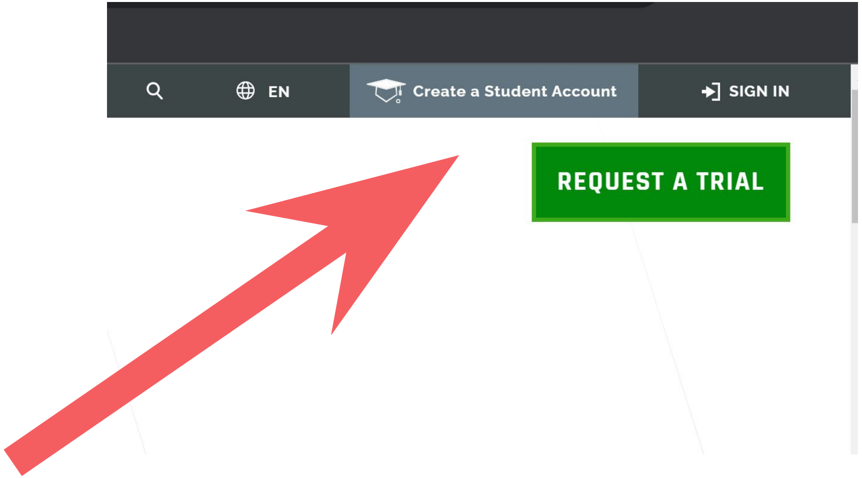





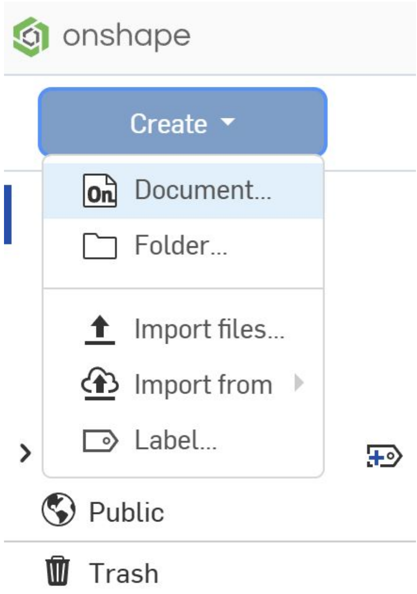
Create an OnShape Account

Create an OnShape account

Go to *onshape.com* and click Create a Student Account in the top right corner

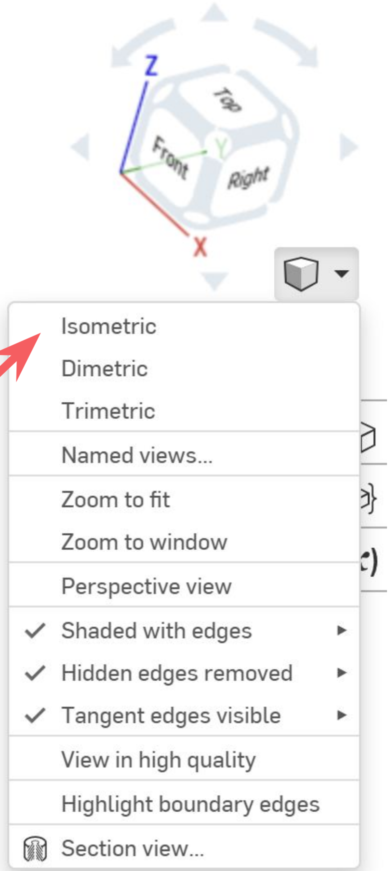


Start a New Project



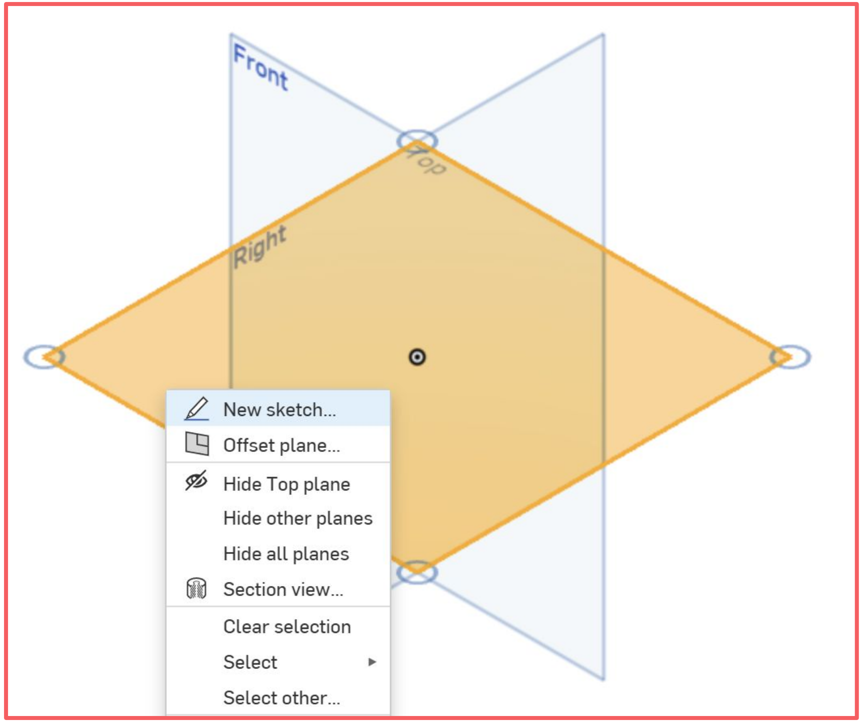
Basic Controls:
Pan = Middle mouse button
Zoom = Mouse wheel
Rotate = Hold right click
Select = Left click
Menu = left click

Default view



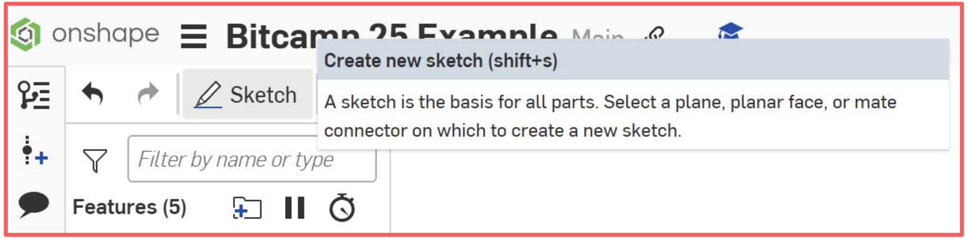
Creating a Sketch

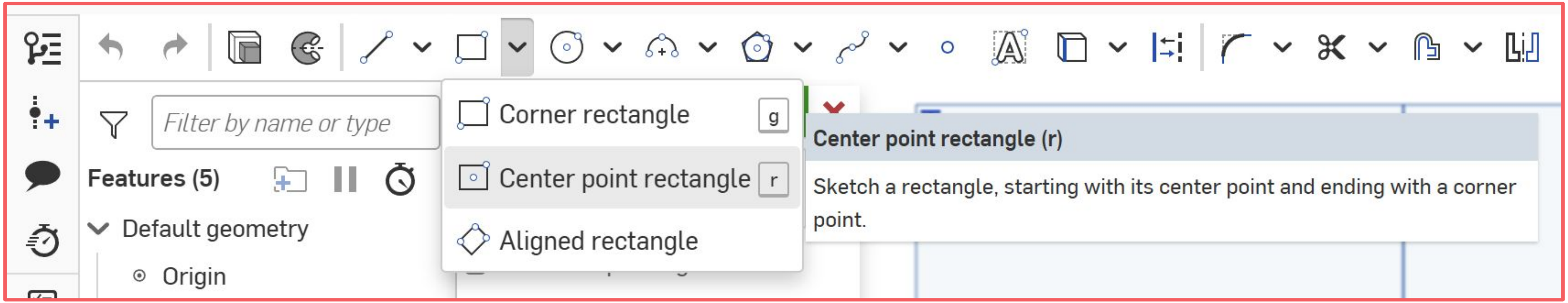
1. Left click the top plane to select it
2. Right click to open menu options
3. Select "New Sketch"



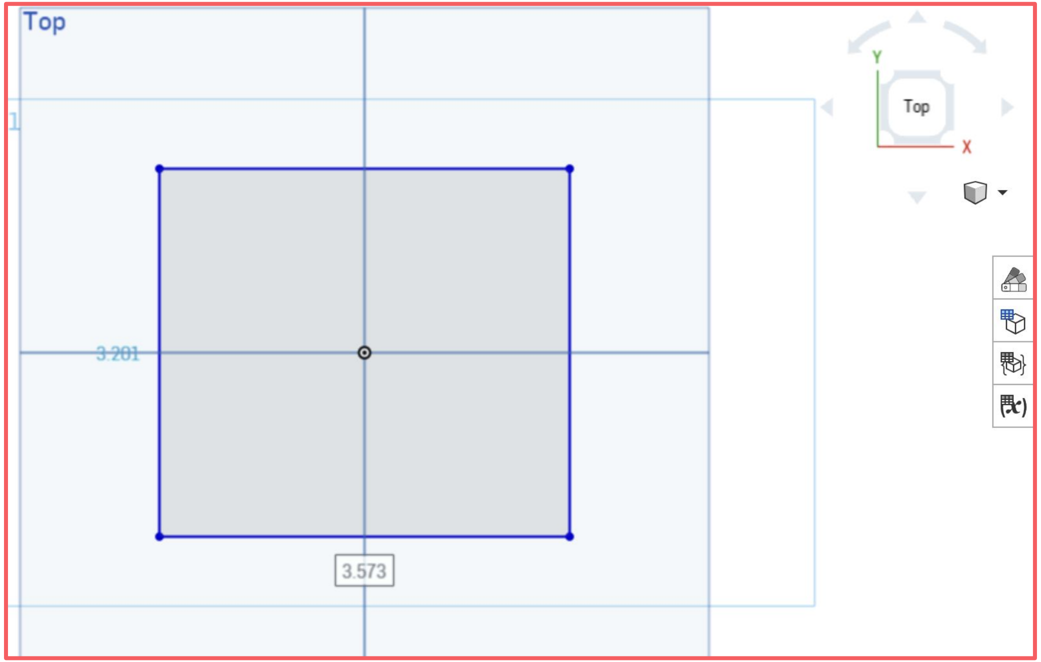
OR

1. Click "New Sketch" in the top left
2. Click on the top plane

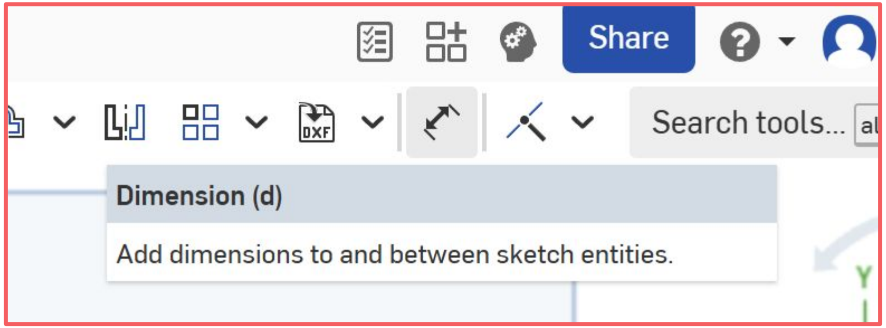


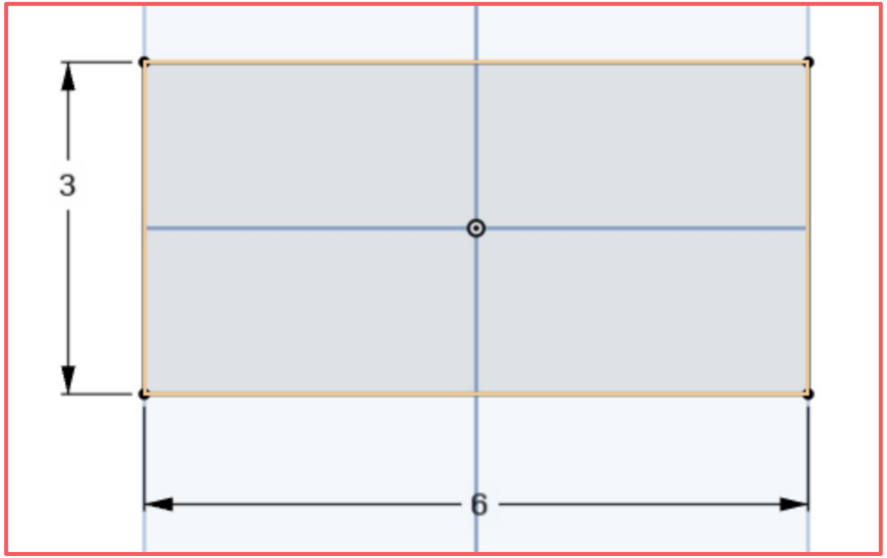


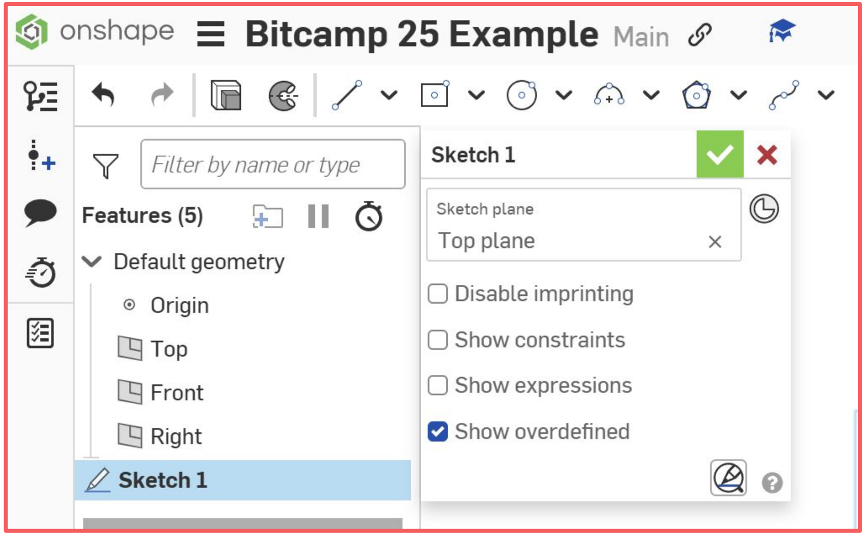
Filter by name or type
Features (5)
Default geometry
Origin
Corner rectangle
g
Center point rectangle
r
Aligned rectangle
Center point rectangle (r)
Sketch a rectangle, starting with its center point and ending with a corner point.

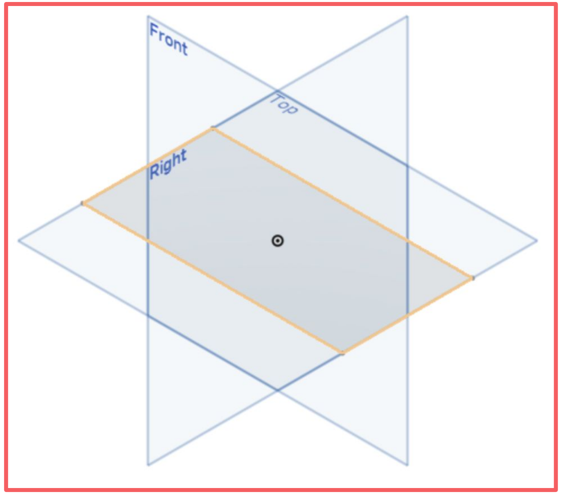


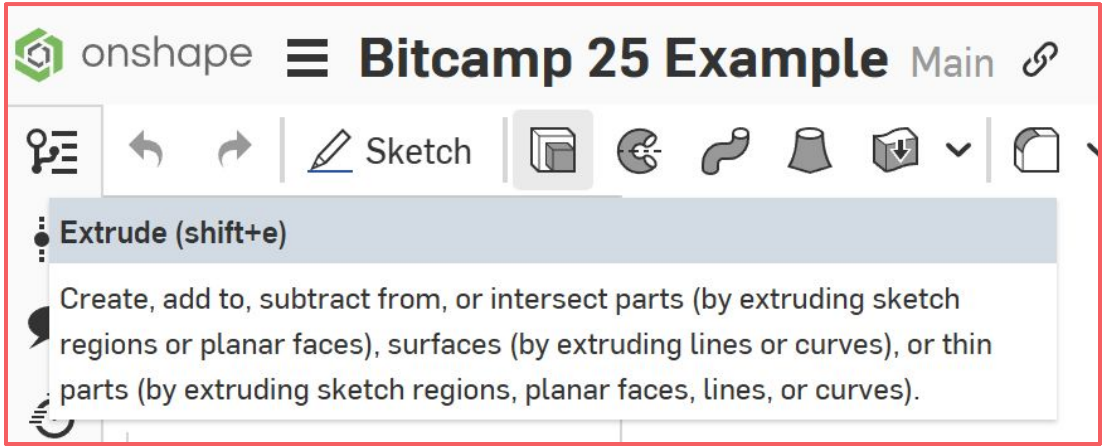
Top
3.201
3.573
Top
Y
X

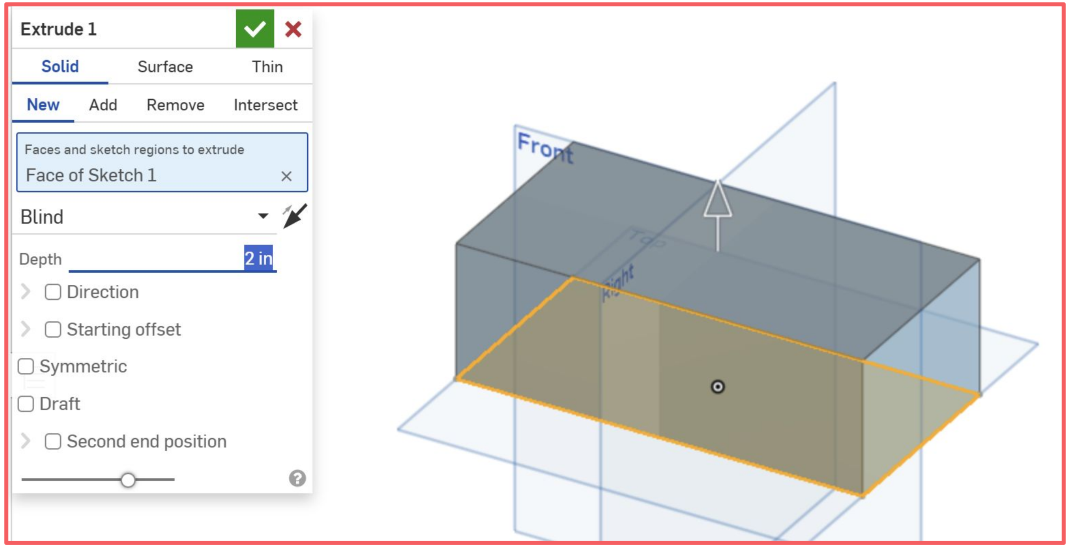

Share
Search tools...
Dimension (d)
Add dimensions to and between sketch entities.

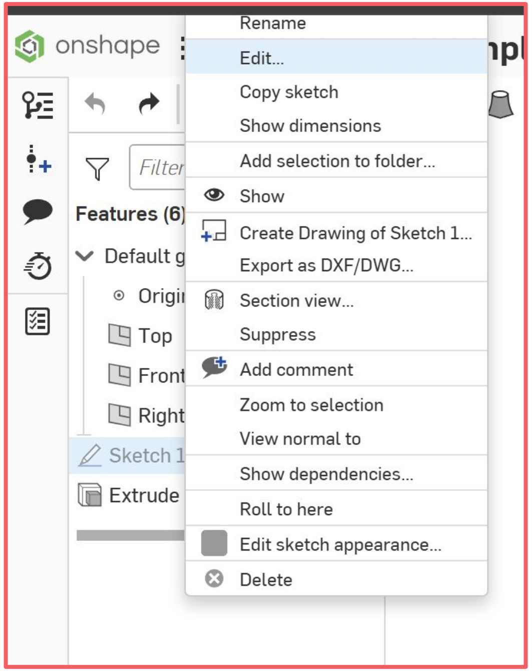

3
6

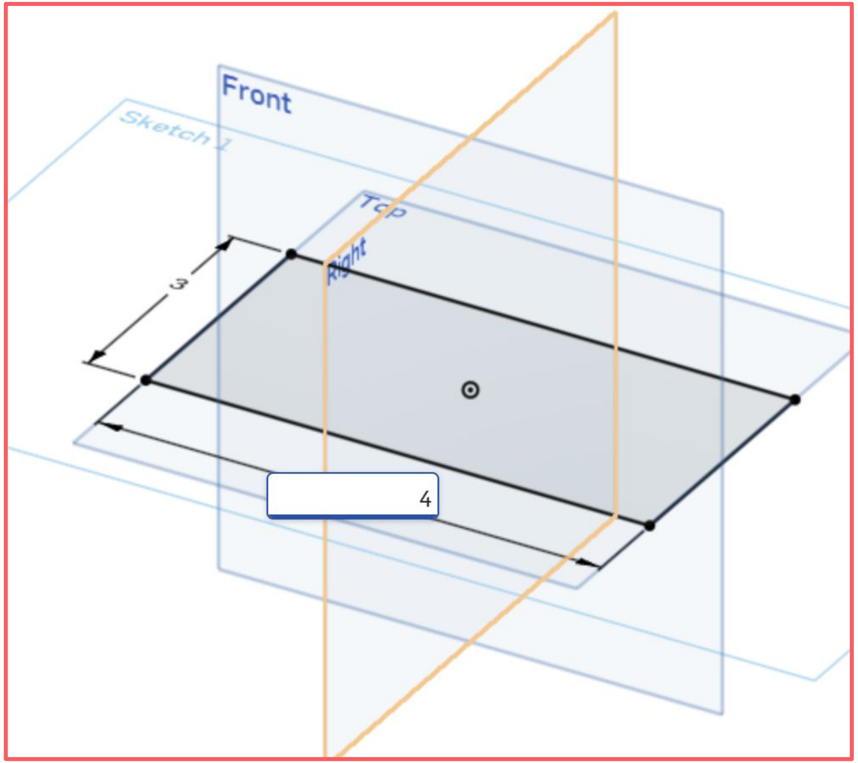

onshape
Bitcamp 25 Example
Main
Filter by name or type
Features (5)
Default geometry
Origin
Top
Front
Right
Sketch 1
Sketch 1
Sketch plane
Top plane
Disable imprinting
Show constraints
Show expressions
Show overdefined

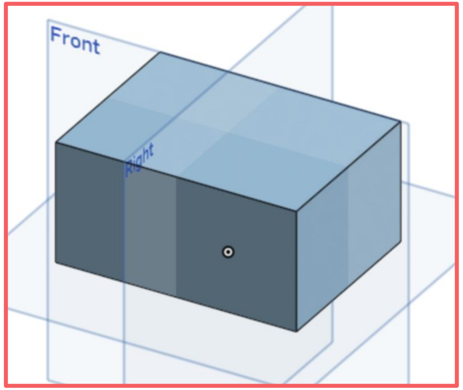

Front
Top
Right


onshape
Bitcamp 25 Example
Main
Sketch
Extrude (shift+e)
Create, add to, subtract from, or intersect parts (by extruding sketch regions or planar faces), surfaces (by extruding lines or curves), or thin parts (by extruding sketch regions, planar faces, lines, or curves).

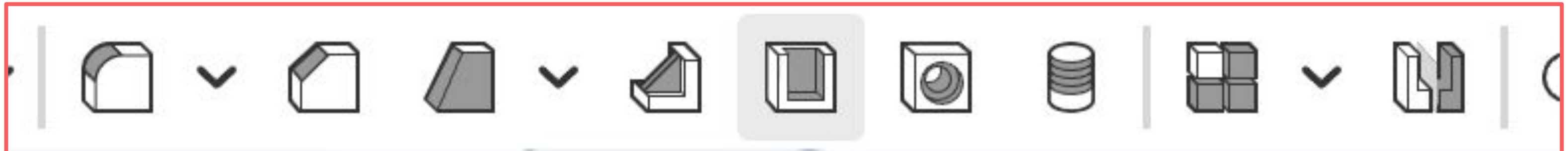

Extrude 1
Solid
Surface
Thin
New
Add
Remove
Intersect
Faces and sketch regions to extrude
Face of Sketch 1
Blind
Depth
2 in
Direction
Starting offset
Symmetric
Draft
Second end position
Front


onshape
Rename
Edit...
Copy sketch
Show dimensions
Add selection to folder...
Show
Create Drawing of Sketch 1...
Export as DXF/DWG...
Section view...
Suppress
Add comment
Zoom to selection
View normal to
Show dependencies...
Roll to here
Edit sketch appearance...
Delete
Filter
Features (6)
Default g
Origi
Top
Front
Right
Sketch 1
Extrude


Front
Sketch 1
Top
Right
3
4

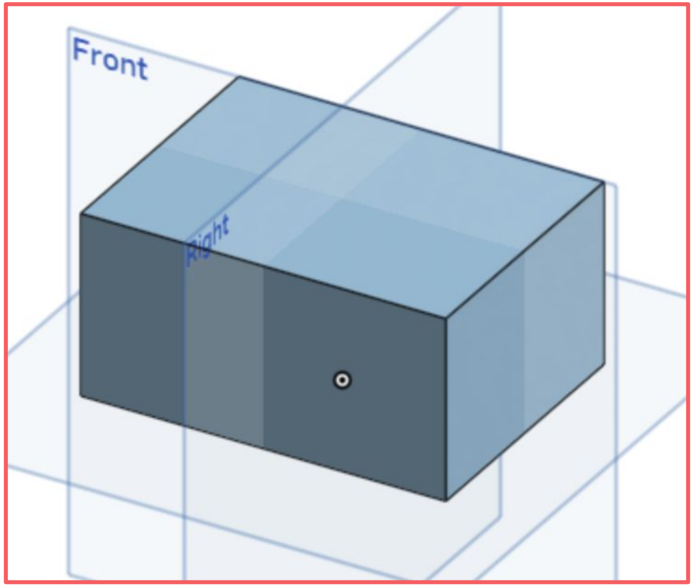

Front
Right

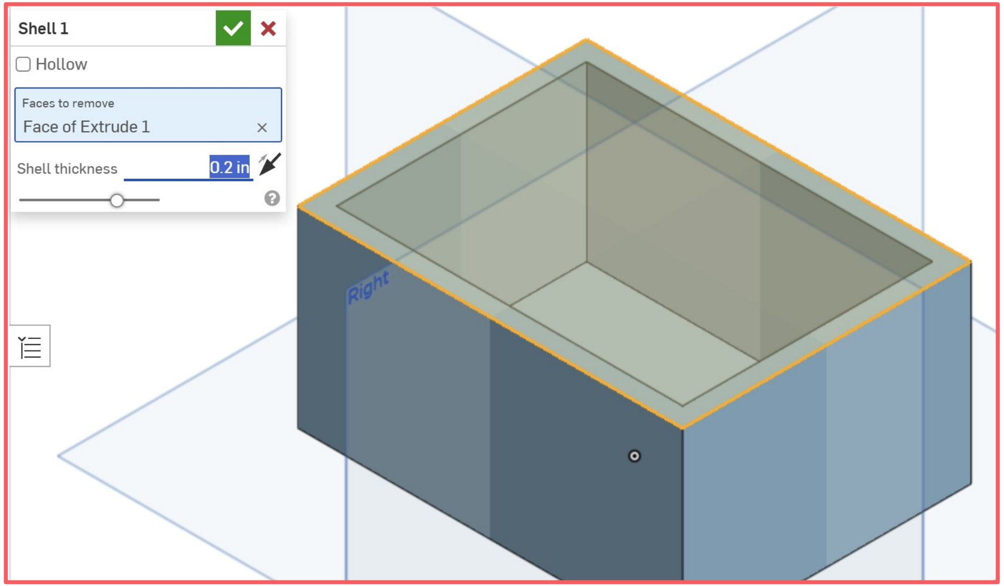
We’re going to model a house:

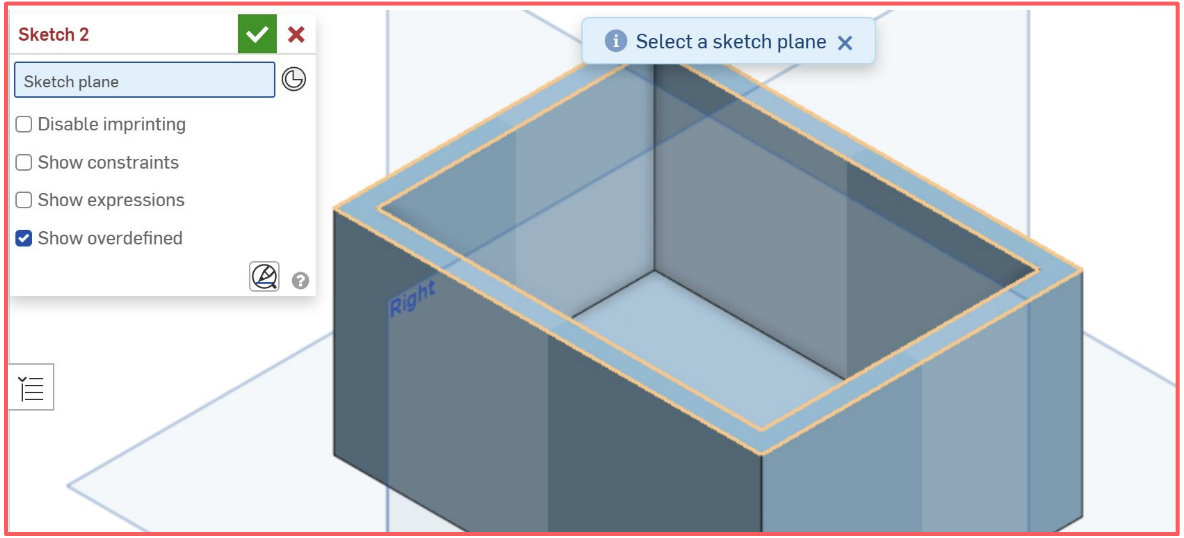


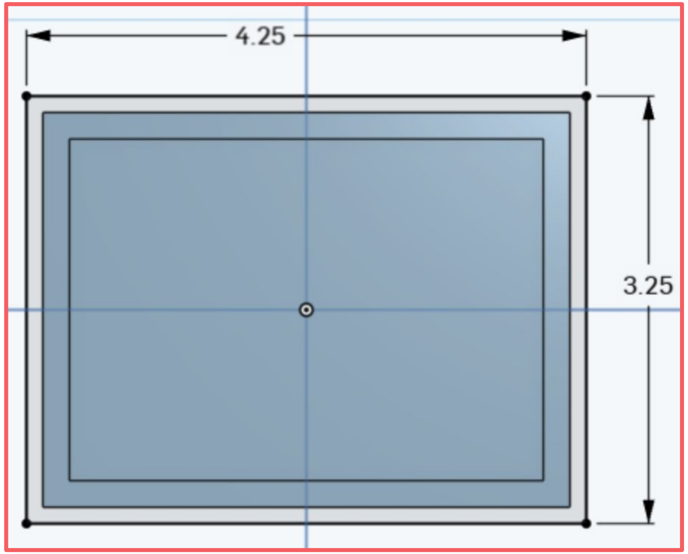
Shell

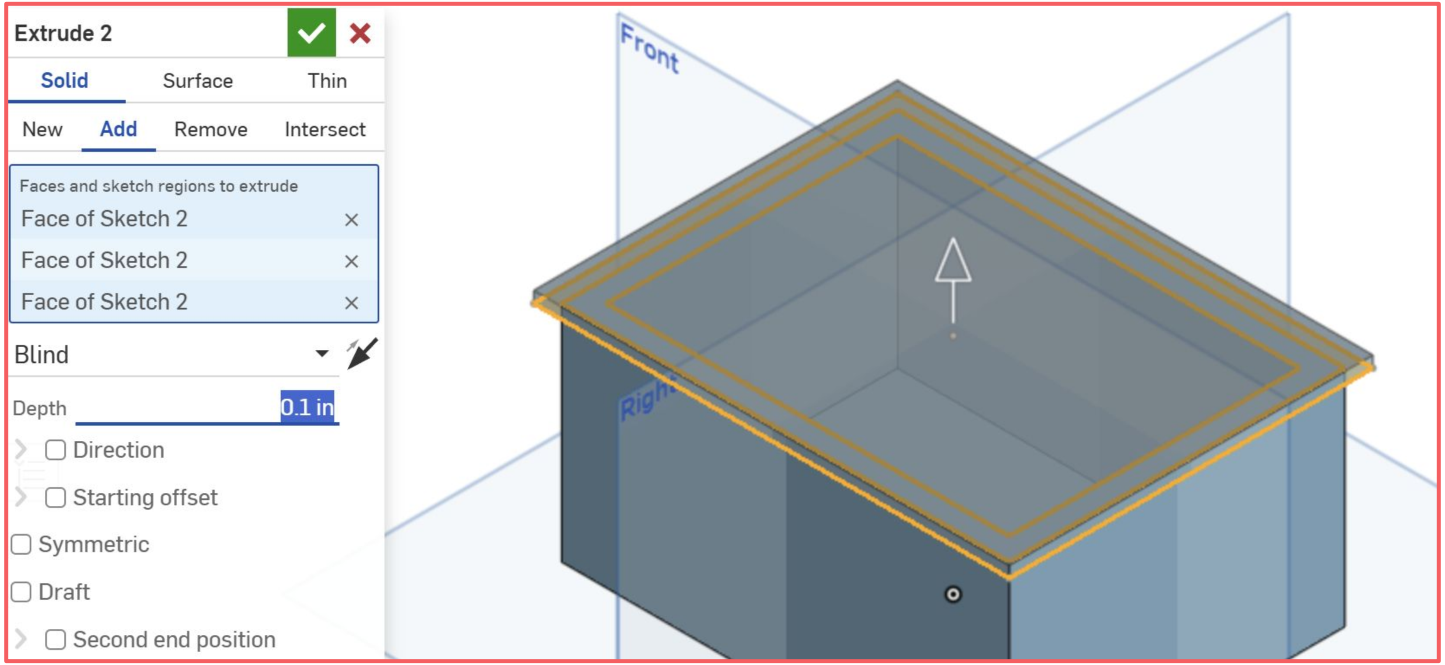
Create a thin-walled part by removing one or more faces. Specify the wall thickness for the remaining faces.



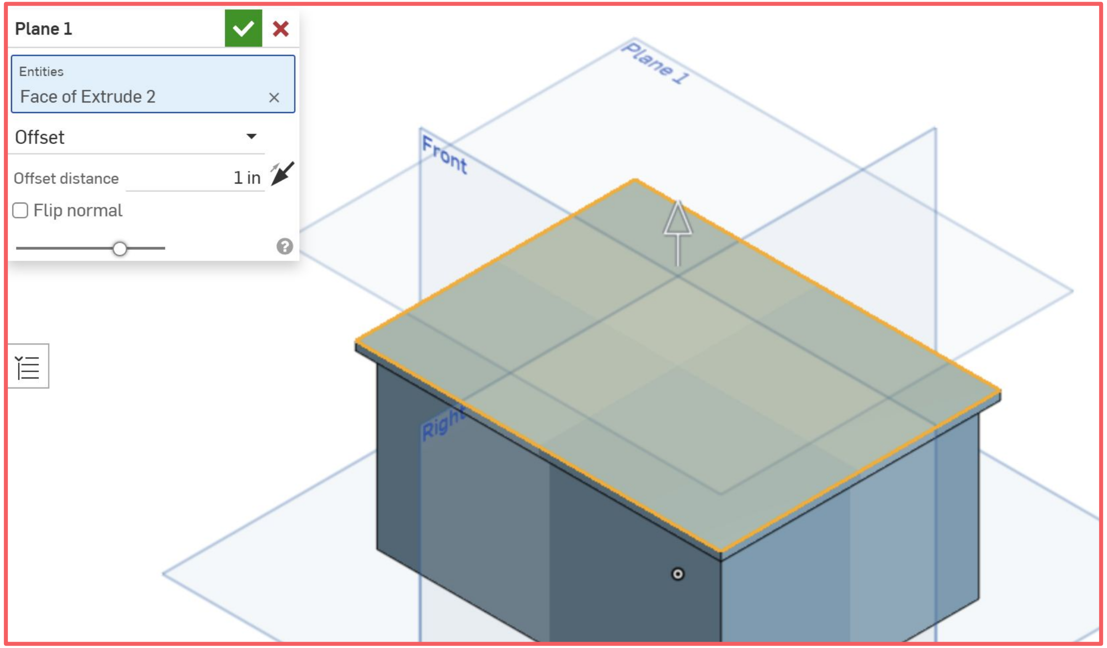


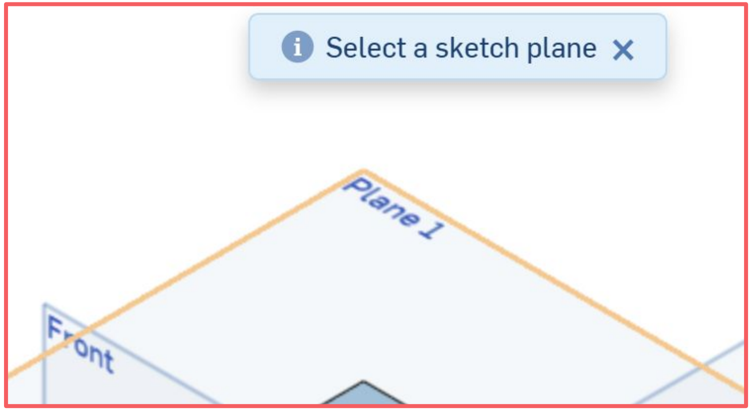

Sketch 2
Sketch plane
Disable imprinting
Show constraints
Show expressions
Show overdefined
Select a sketch plane
Right

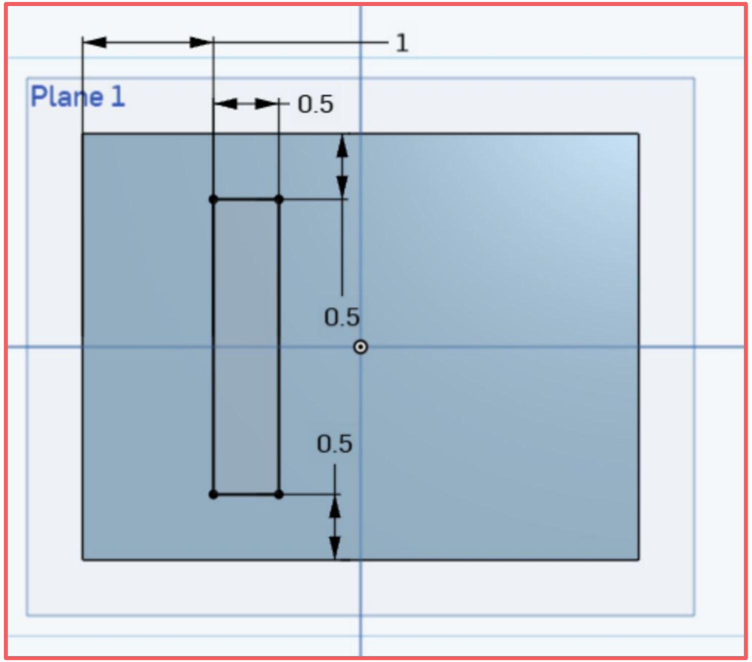

4.25
3.25

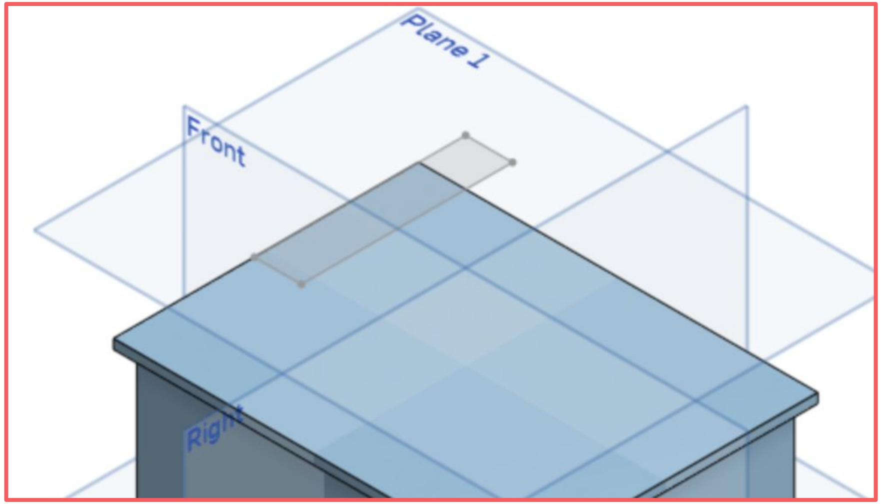

Extrude 2
Solid
Surface
Thin
New
Add
Remove
Intersect
Faces and sketch regions to extrude
Face of Sketch 2
Face of Sketch 2
Face of Sketch 2
Blind
Depth
0.1 in
Direction
Starting offset
Symmetric
Draft
Second end position
Front
Right

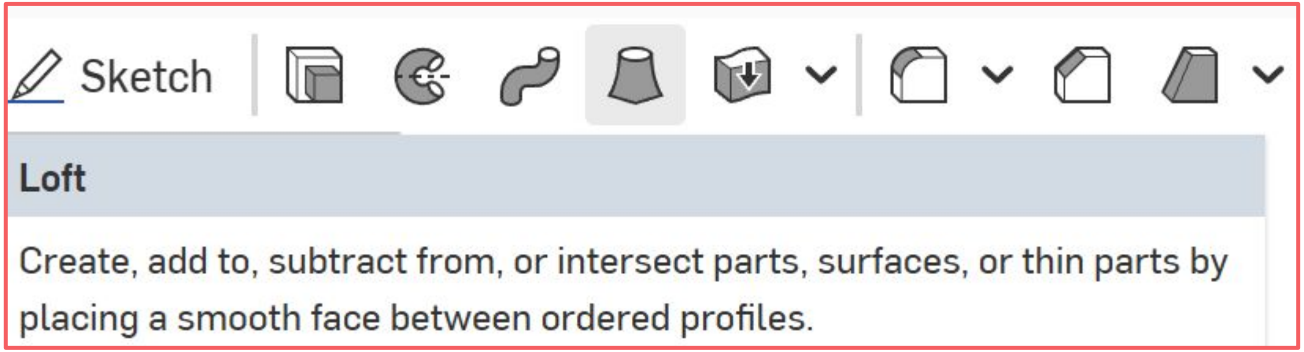

Plane
Create a new construction plane by referencing existing geometry.

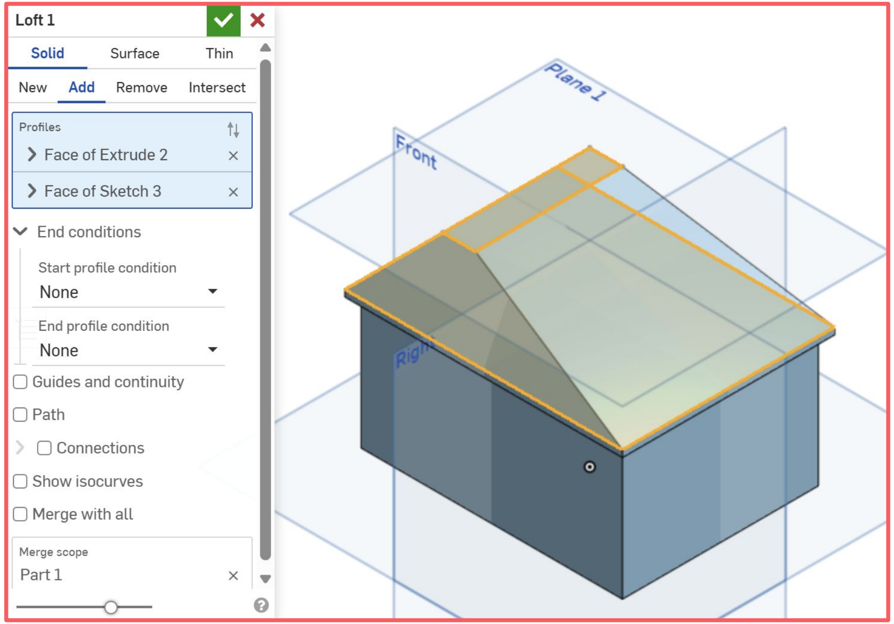

Plane 1
Entities
Face of Extrude 2
Offset
Offset distance
1 in
Flip normal
Plane 1
Front
Right

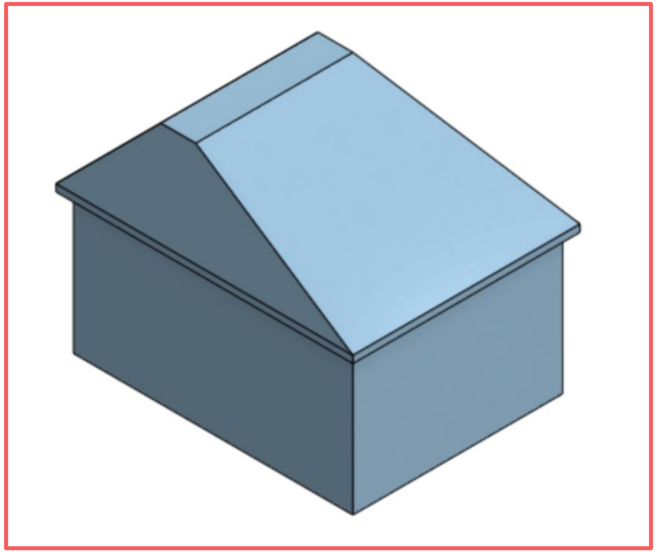

Select a sketch plane
Plane 1
Front

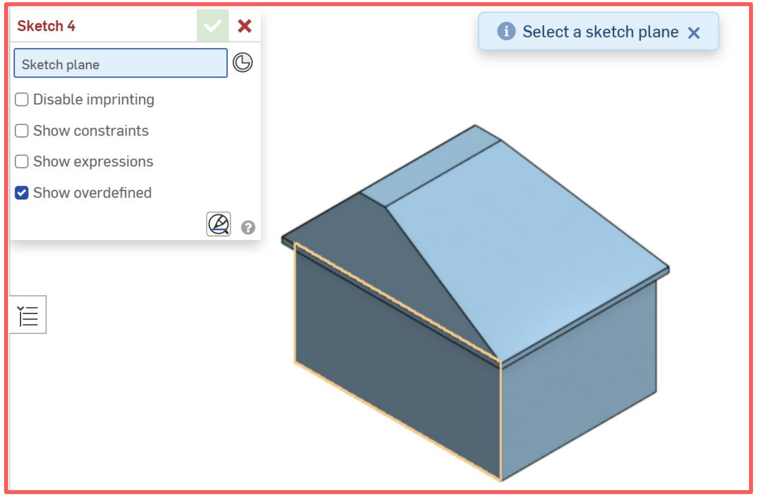

1
Plane 1
0.5
0.5
0.5

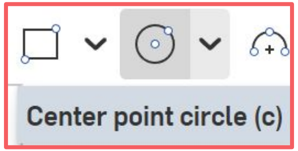

Plane 1
Front
Right

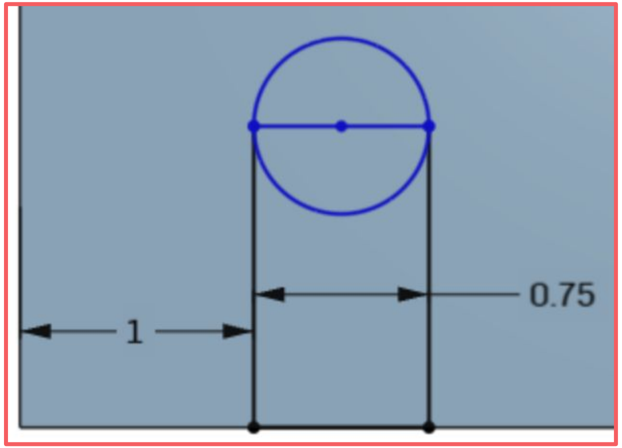

Sketch
Loft
Create, add to, subtract from, or intersect parts, surfaces, or thin parts by placing a smooth face between ordered profiles.

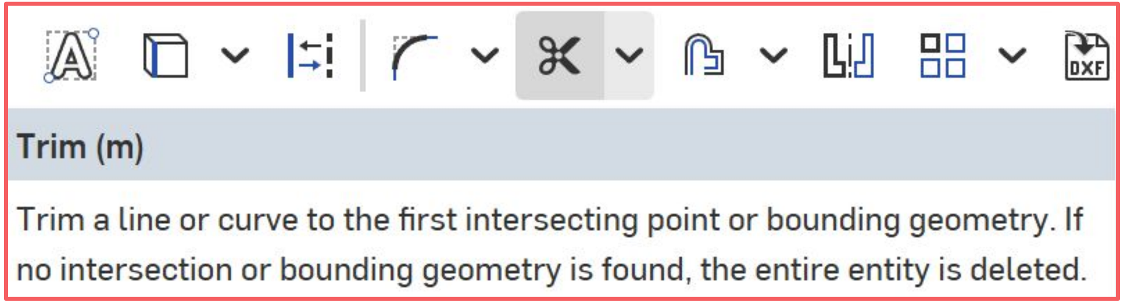

Loft 1
Solid
Surface
Thin
New
Add
Remove
Intersect
Profiles
Face of Extrude 2
Face of Sketch 3
End conditions
Start profile condition
None
End profile condition
None
Guides and continuity
Path
Connections
Show isocurves
Merge with all
Merge scope
Part 1
Plane 1
Front

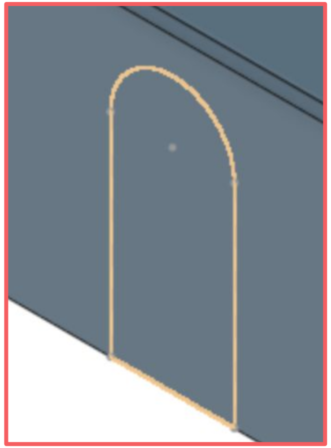


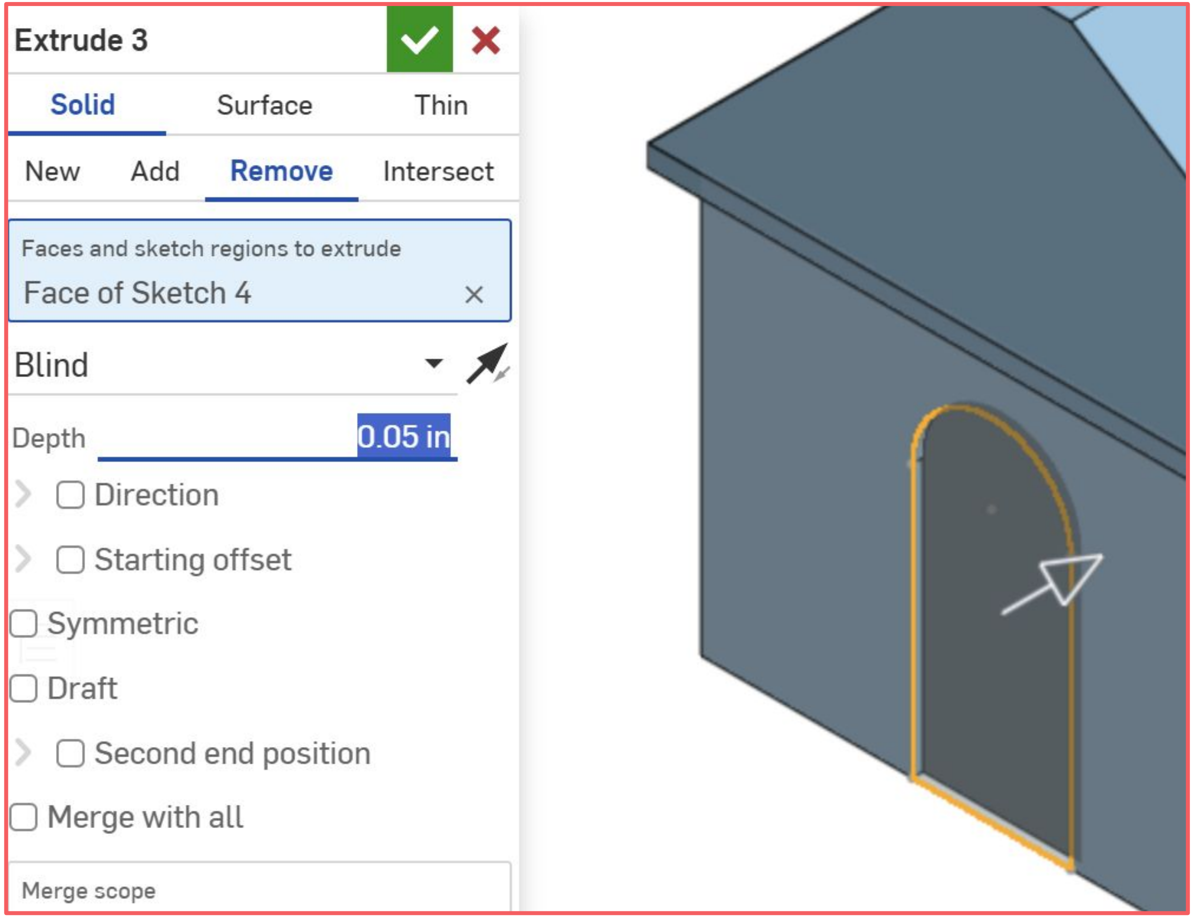

Sketch 4
Sketch plane
Disable imprinting
Show constraints
Show expressions
Show overdefined
Select a sketch plane

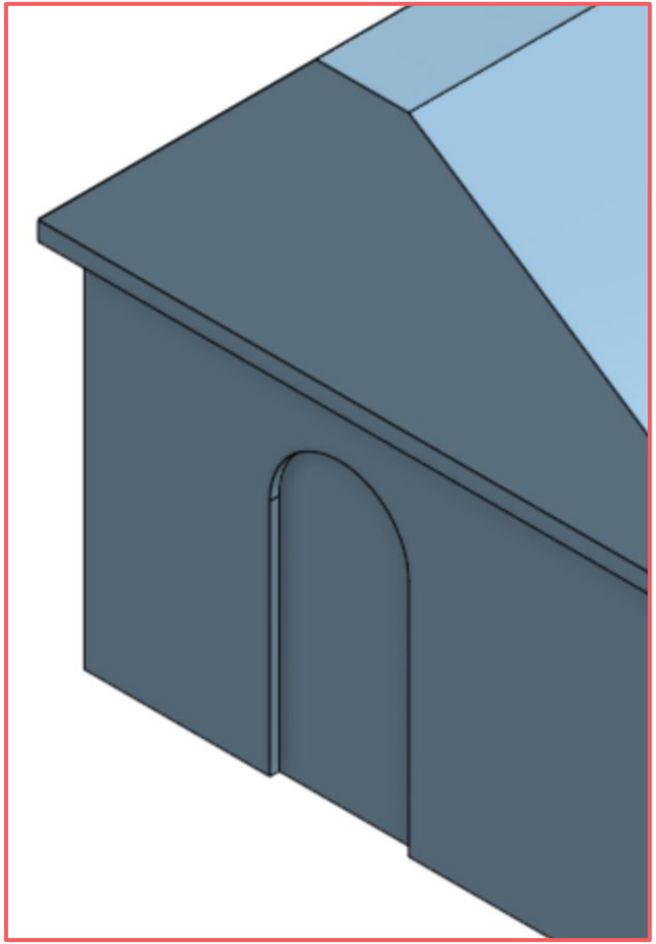

Center point circle (c)

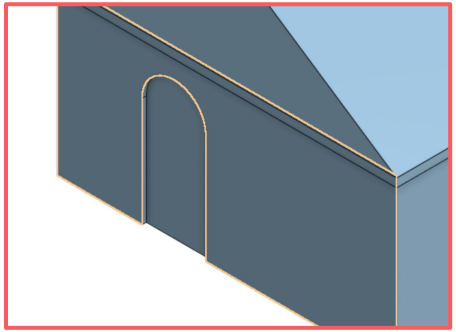

0.75
1


DXF
Trim (m)
Trim a line or curve to the first intersecting point or bounding geometry. If no intersection or bounding geometry is found, the entire entity is deleted.

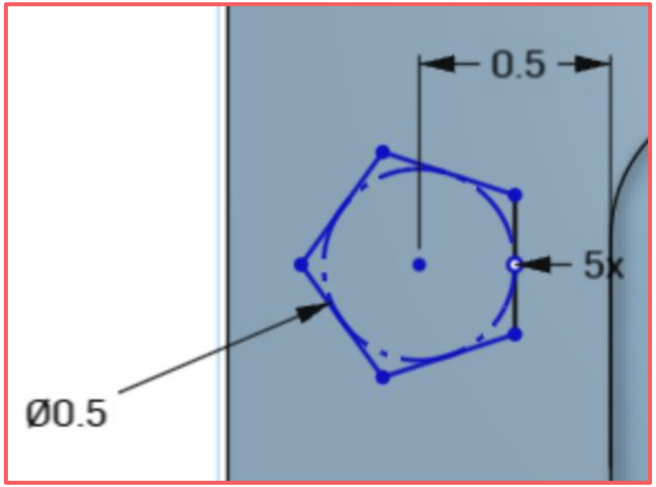


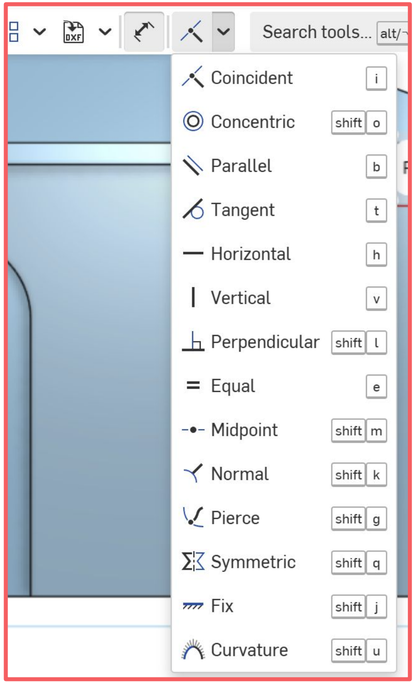

Extrude 3
Solid
Surface
Thin
New
Add
Remove
Intersect
Faces and sketch regions to extrude
Face of Sketch 4
Blind
Depth
0.05 in
Direction
Starting offset
Symmetric
Draft
Second end position
Merge with all
Merge scope

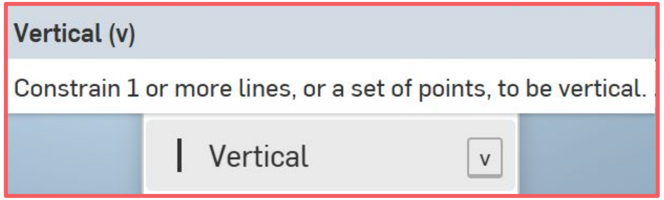


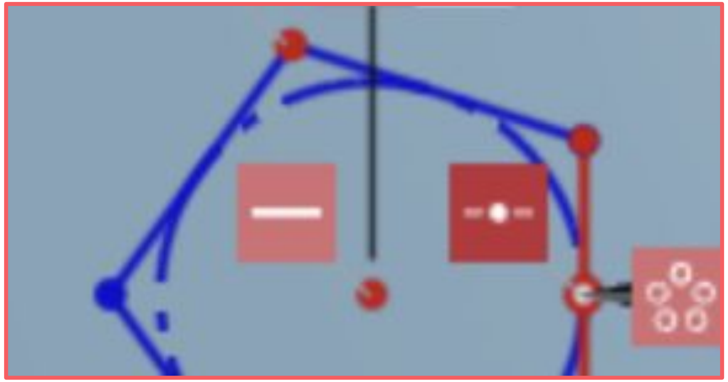


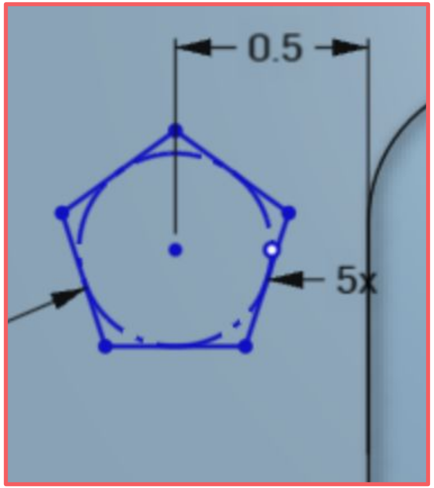

Inscribed polygon
Sketch a polygon defined on an inscribed circle.

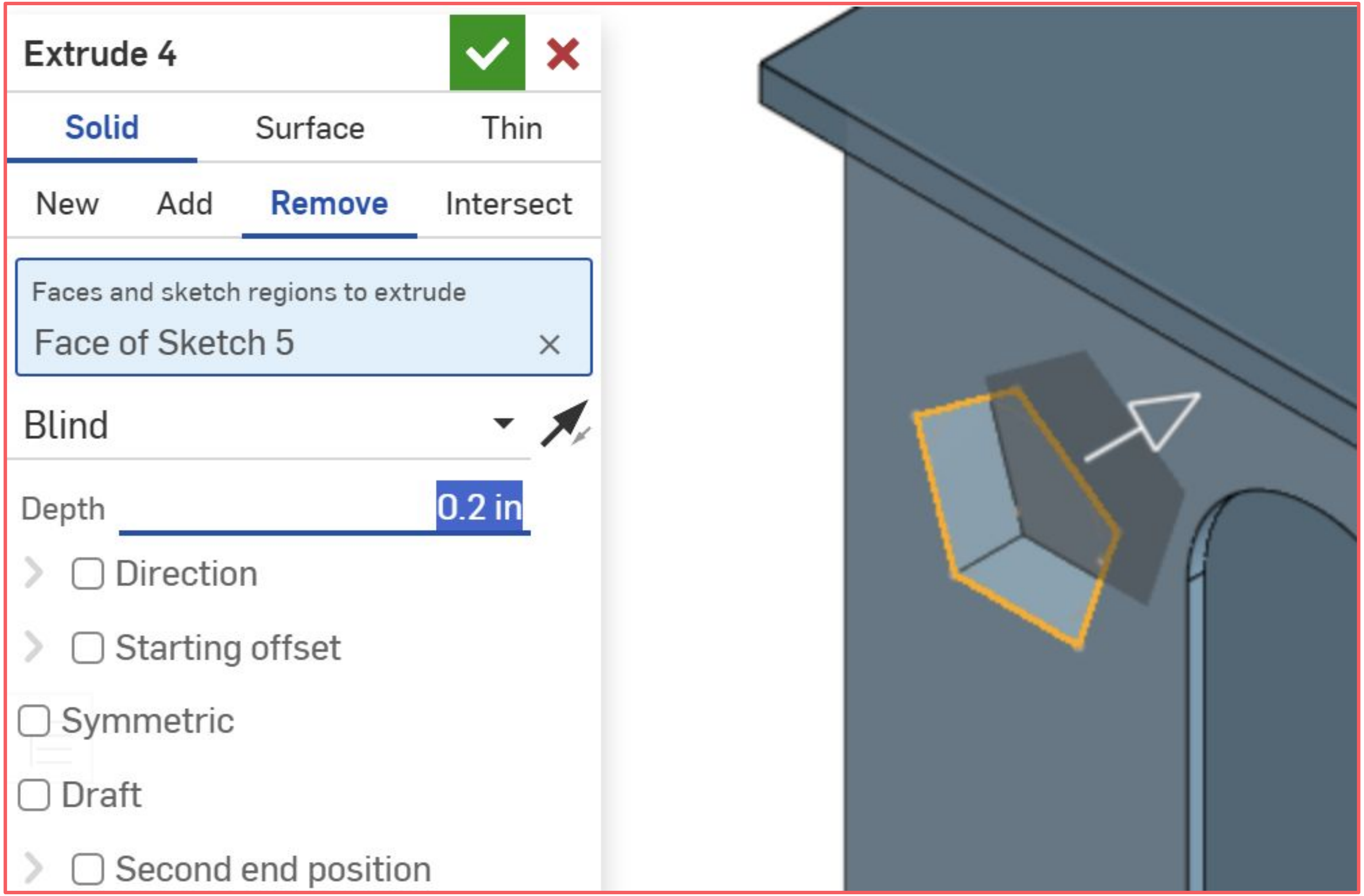

0.5
5x
Ø0.5

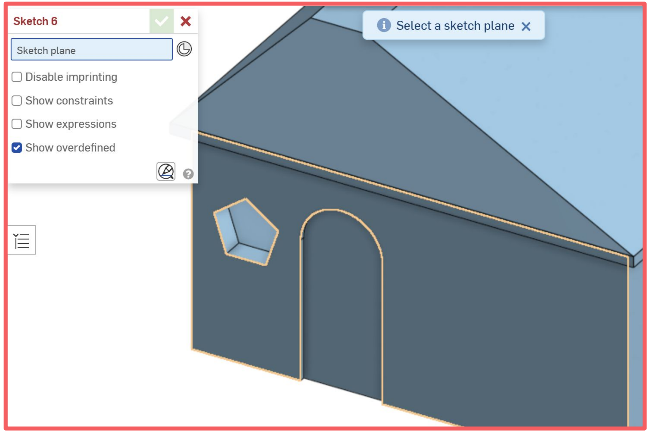

Search tools...
Coincident i
Concentric shift o
Parallel b
Tangent t
Horizontal h
Vertical v
Perpendicular shift l
Equal e
Midpoint shift m
Normal shift k
Pierce shift g
Symmetric shift q
Fix shift j
Curvature shift u

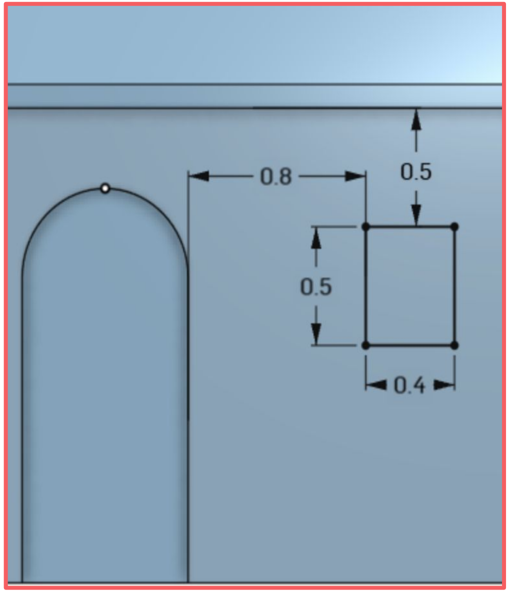

Vertical (v)
Constrain 1 or more lines, or a set of points, to be vertical.
Vertical v




0.5
5x

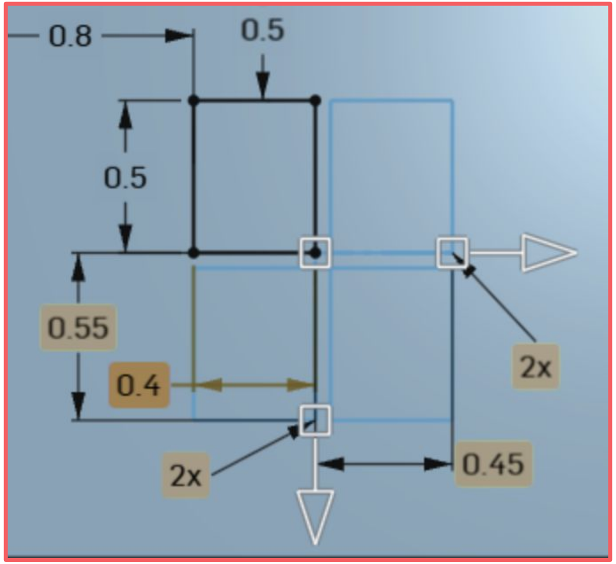

Extrude 4
Solid
Surface
Thin
New
Add
Remove
Intersect
Faces and sketch regions to extrude
Face of Sketch 5
Blind
Depth
0.2 in
Direction
Starting offset
Symmetric
Draft
Second end position

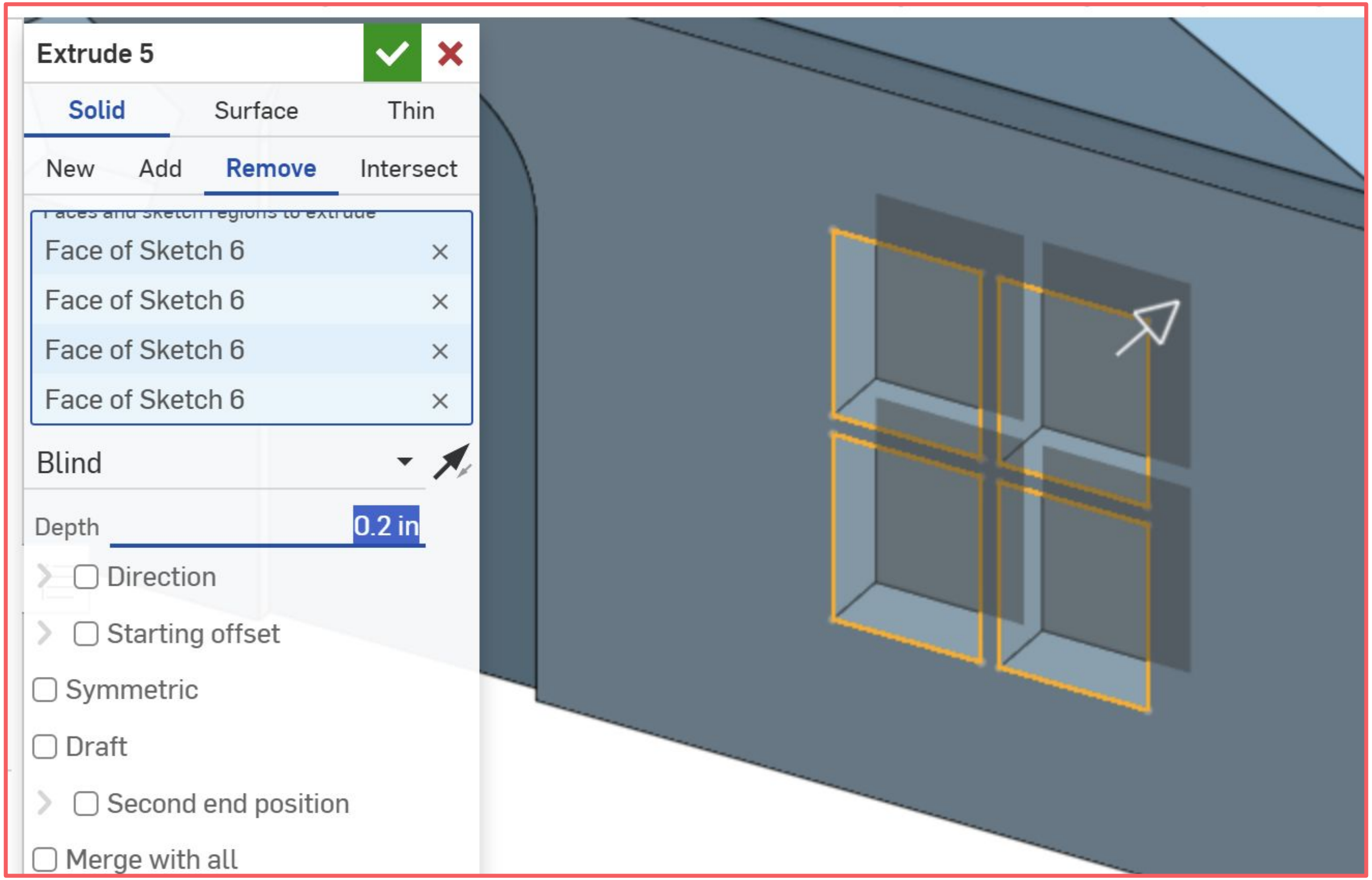


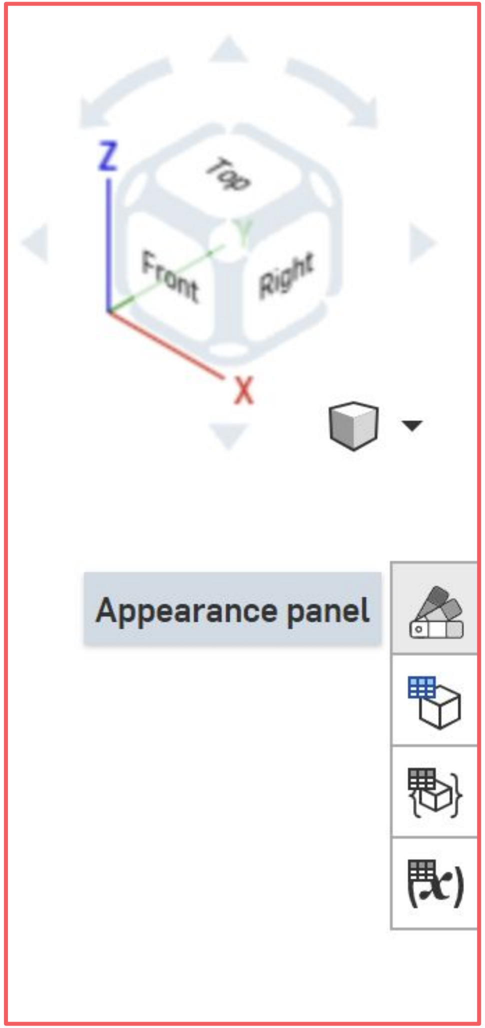


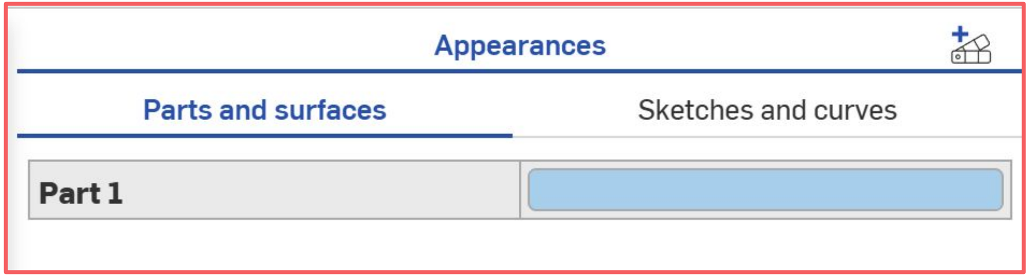
Linear pattern

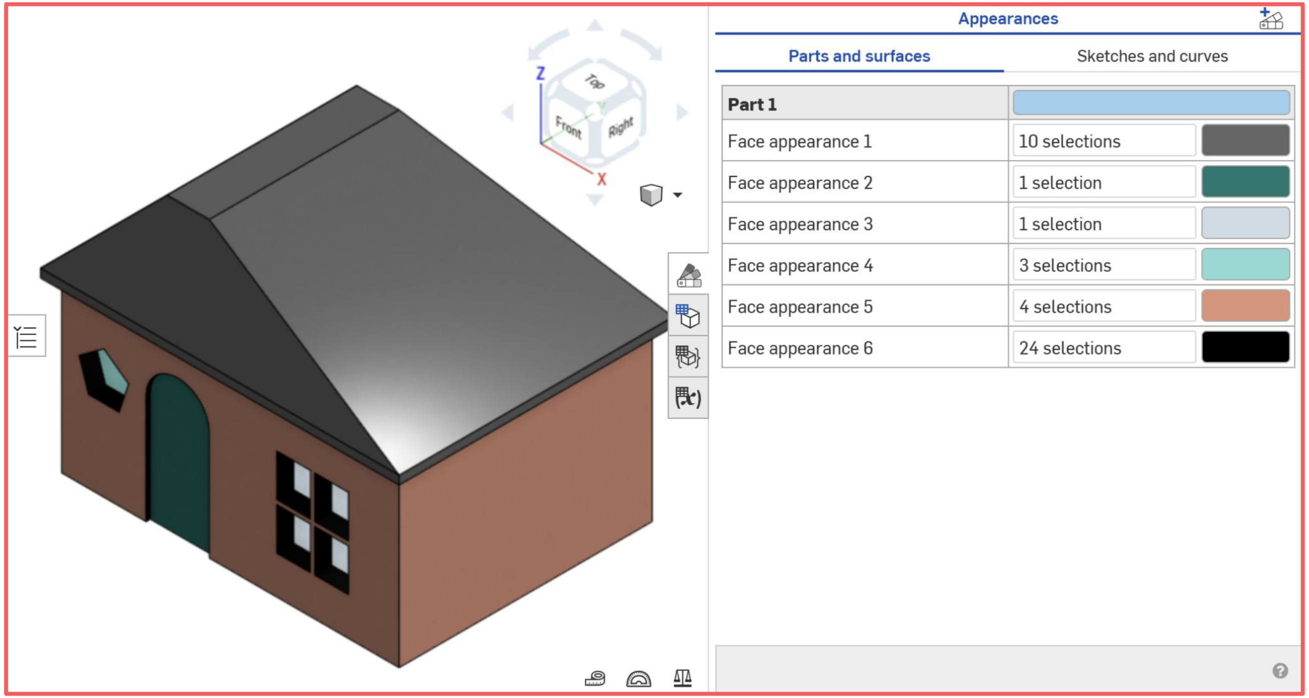
Create multiple instances of sketch entities uniformly in 1 or 2 linear directions. Specify the distance between instances and the number of instances.

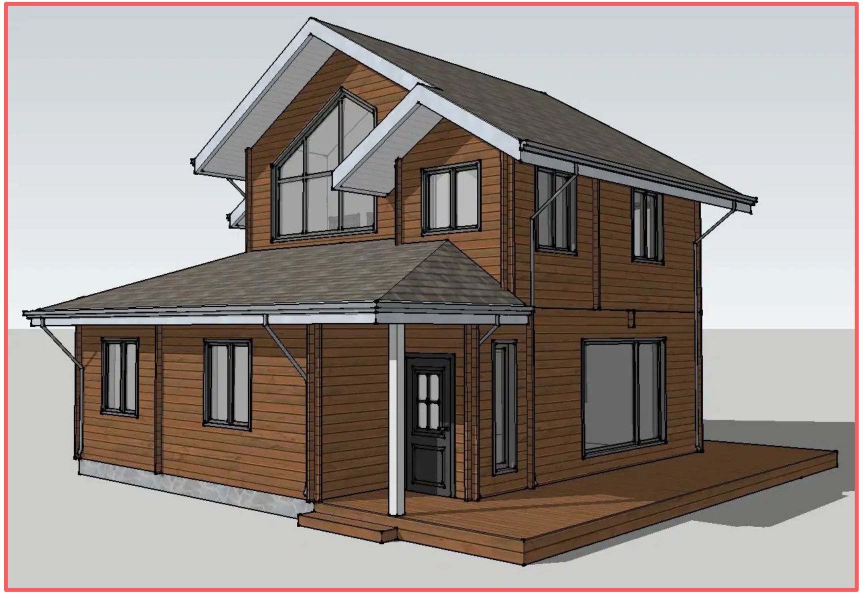



Extrude 5
Solid
Surface
Thin
New
Add
Remove
Intersect
Face of Sketch 6
Face of Sketch 6
Face of Sketch 6
Face of Sketch 6
Blind
Depth
0.2 in
Direction
Starting offset
Symmetric
Draft
Second end position
Merge with all


Z
Top
Front
Right
X
Appearance panel


Appearances
Parts and surfaces
Sketches and curves
Part 1


Z
Top
Front
Right
X
Appearances
Parts and surfaces
Sketches and curves
Part 1
Face appearance 1
10 selections
Face appearance 2
1 selection
Face appearance 3
1 selection
Face appearance 4
3 selections
Face appearance 5
4 selections
Face appearance 6
24 selections



By Vaishali on cadbull.com

By u/AutoDefenestrator273 on reddit.com

Thank you for coming,
we’d love to hear your feedback :)

https://forms.gle/NeYccXwrwdHXp2J1A